Roll No. ....

5th Sem/ ECE, Automation &
Robotics, ECE (For Speech & Hearing/Impaired)
Subject : PLC & SCADA

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 A Nibble Represents

a) 4 Bits b) 8 Bits
c) 16 Bits d) 2 Bits

Q.2 The acronym PLC stands for :

a) Pressure Load Control
b) Programmable Logic Controller
c) Pneumatic Logic Capstan
d) Pressure Loss Chamber

Q.3 In P.L.C file is a block of ______.

a) Data b) Memory
c) Instructions d) None of above

Q.4 An AND function implemented in ladder logic uses

a) Normally closed contacts in series.
b) Normally open contacts in series
c) Normally open contacts in parallel
d) Normally closed contacts in parallel

Q.5 Full form of SCADA is :

a) Superior Control & Direct Access

b) Supervisory Control & Direct Access

c) Supervisory Control & Data Acquisition

d) superior Control & Data Acquisition

Q.6 In SCADA system, Tags can be of following types

a) Machine Tag b) Cloth Tag

c) Computer Tag d) Memory Tag

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write the full form of SFC.

Q.8 ROM Stands for _____.

Q.9 A byte consists of _____ bits.

Q.10 NAND gate is a Universal Gate (T/F)

Q.11 _____ instructions will allow the PLC to bypass some Ladder Logic Instructions.

Q.12 Define Rungs in PLC.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What is tag & explain its types in brief.

Q.14 Explain the uses of Sliders in SCADA.

Q.15 Describe Timer Data File of P.L.C.

Q.16 Describe Data Logging in Context of SCADA.

Q.17 Write any four difference between Relay & P.L.C.

Q.18 Describe the working of Relay.

Q.19 Explain Output Data File Structure.

Q.20 Write any four comparison Instruction.

Q.21 Describe the Function of SHIFT register Instructions.

Q.22 Explain the Logic Expression, symbol & Truth table of XOR & XNOR gates.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the steps used in Creation of a SCADA Project.

Q.24 Explain Different Sequencer Instructions of PLC.

Q.25 With the help of block diagram Explain Architecture of P.L.C.